



Important Points for IoT Module

Hardware:

1. This Program focuses on the software side including cloud usage.
2. So, the contents will be primarily focused on the software side.
3. We will not be discussing/showcasing any hardware devices.
4. We'll use software simulator instead of actual hardware
5. For those interested in exploring and understanding more hardware, please refer below.
 - a. Below Hardware Section will help you to use the raspberry device and temperature sensor.
 - b. The document is focused on the OS set up on the Raspberry.
 - c. It also covers a simple temperature logger experiment using LM75A (type of a temperature sensor).

Note - This is completely optional.



Hardware

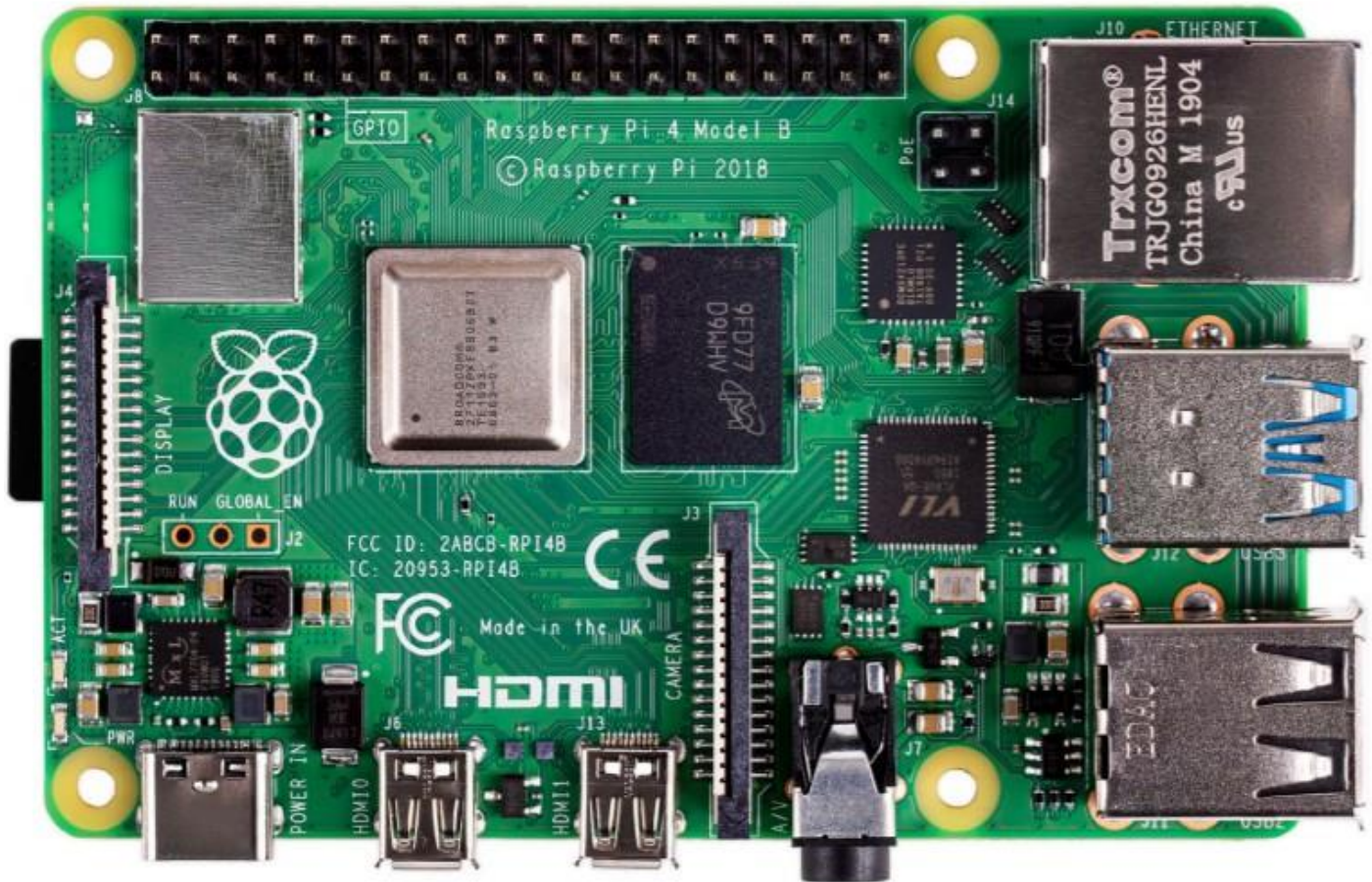
- Raspberry Pi
 - Features
 - Documentation
 - Setup and Installation of OS
 - Running it for the first time
 - LM75A Temperature sensor
 - Description
 - Features
 - Applications
 - Documentation
 - Temperature logger experiment
 - Additional hardware requirement
 - Circuit Connection
 - Python Code
 - Output

Raspberry Pi 4

Raspberry Pi is the name of a series of powerful single-board computers. These devices are developed by Raspberry Pi Foundation which aims to educate people in computing and create easier access to computing education. The very first Raspberry Pi was launched in 2012 with a single core 700MHz CPU and just 256MB RAM. There have been several iterations released since then and currently we have Raspberry Pi 4 as the latest version.

The Raspberry Pi is a cheap computer that runs on Linux OS. Raspberry Pi's provide GPIOs (general purpose input/output) pins that allow you to connect and control electronic components for physical computing and explore the Internet of Things (IoT). All over the world people are using Raspberry Pi devices to learn/develop programming skills, build hardware projects, do home automation, and even use them in industrial applications.

This section contains the basic technical details of the Raspberry Pi 4. We will share the official documentation which you can go through for deeper understanding.



Features

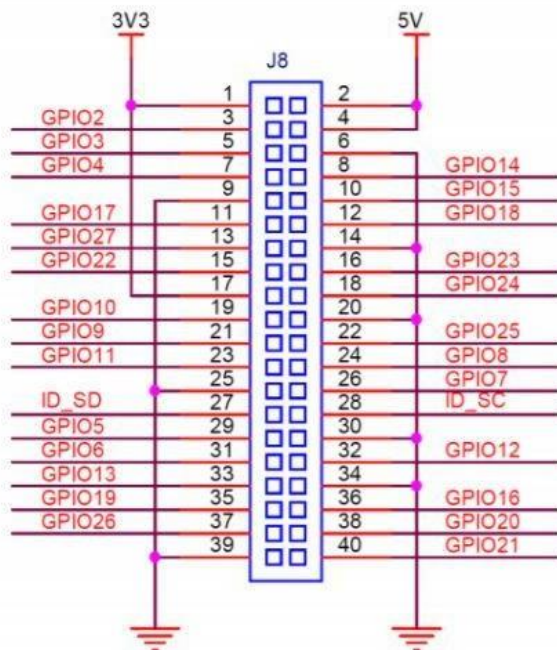
Hardware & Interfaces

- Broadcom BCM2711, Quad core Cortex-A72 (ARM v8) 64-bit SoC @ 1.5GHz
- 2GB, 4GB or 8GB LPDDR4-3200 SDRAM variants
- 2.4 GHz and 5.0 GHz IEEE 802.11ac wireless, Bluetooth 5.0, BLE
- Gigabit Ethernet
- 2 USB 3.0 ports; 2 USB 2.0 ports.
- Raspberry Pi standard 40 pin GPIO header (fully backwards compatible with previous boards)
- 2 × micro-HDMI ports (up to 4kp60 supported)
- 2-lane MIPI DSI display port
- 2-lane MIPI CSI camera port



- 4-pole stereo audio and composite video port
- H.265 (4kp60 decode), H264 (1080p60 decode, 1080p30 encode)
- OpenGL ES 3.0 graphics
- Micro-SD card slot for loading operating system and data storage
- 5V DC via USB-C connector (minimum 3A*)
- 5V DC via GPIO header (minimum 3A)

- Power over Ethernet (PoE) enabled (requires separate PoE HAT)
- **GPIO Interface:** These pins are designed to be used by straightforward software controlled input and output (with programmable pulls) also GPIO pins can be switched (multiplexed) into various other modes such as I2C, UART and SPI. This gives users much more flexibility when attaching any add-on hardware.



ID_SD and ID_SC PINS:

These pins are reserved for HAT ID EEPROM.

At boot time this I2C interface will be interrogated to look for an EEPROM that identifies the attached board and allows automatic setup of the GPIOs (and optionally, Linux drivers).

DO NOT USE these pins for anything other than attaching an I2C ID EEPROM. Leave unconnected if ID EEPROM not required.

Software

- ARMv8 Instruction Set
- Mature Linux software stack
- Recent Linux kernel support
- Availability of GPU functions using standard APIs

Documentation

- [Raspberry Pi Documentation](#)

Installation & Setup

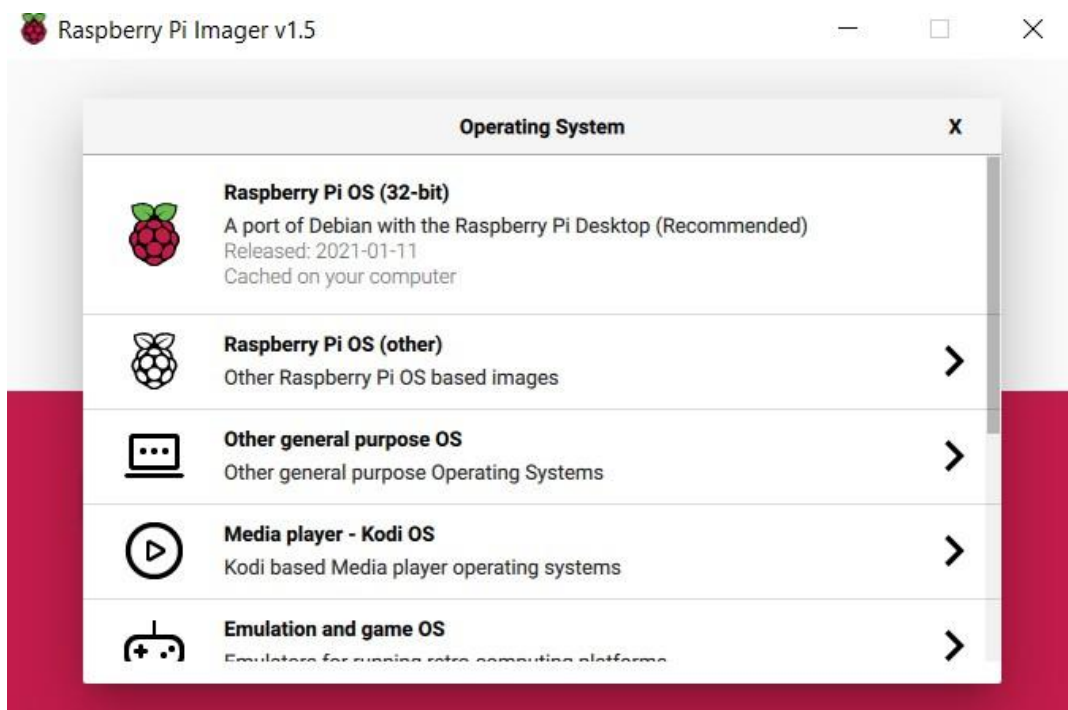
This resource explains how to install a Raspberry Pi operating system image on an SD card. You will need another computer with an SD card reader to install the image.

There is a SD card writing tool available on the official Raspberry Pi website. The tool is compatible with different operating systems such as Mac OS, Ubuntu >=18.04 and Windows 10. This is the easiest option for users as it will download the image and install it automatically to the SD card.

- Download the latest version of [Raspberry Pi Imager](#) and install it.
- Connect an SD card reader with the SD card inside to the laptop.
- Open the Raspberry Pi Imager



- Click on choose OS. Choose Raspberry Pi OS (32 bit)



- Choose the appropriate SD card listed in the drop down menu.
- Review your selections and click 'WRITE' to begin writing data to the SD
- Note: if you are using the Raspberry Pi Imager on Windows 10 with Controlled Folder Access enabled, you will need to explicitly allow the Raspberry Pi Imager permission to write the SD card. If this is not done, Raspberry Pi Imager will fail with a "failed to write" error.



card.

Boot your new OS

- You can now insert the SD card into the Raspberry Pi and power it up.
- Connect the raspberry pi to the monitor using HDMI cable.
- For the official Raspberry Pi OS, if you need to manually log in, the default user name is pi, with password raspberry. Remember the default keyboard layout is set to UK.
- Change the default password straight away to ensure your device is secure.

Doing the headless Setup

When you are booting the raspberry pi for the first time and you do have the monitor then follow the below mentioned steps:

- Connect the raspberry device to the monitor.
- Do the initial setup such as choose the language, timezone etc and setup the Wireless connection.
- Your raspberry device is ready and available to use.

However if you do not have the monitor then you need to do the headless setup which will enable you to use your laptop's monitor and keyboard to use the raspberry pi GUI.

Please follow the below mentioned steps to complete this process.

- Connect your raspberry pi device through a LAN cable. Make sure that the Internet is working for that network.
- Once the LAN cable is connected procure the IP address of the Raspberry device.

Connecting via SSH

After you have the Pi connected to your network or directly to your PC, you'll need to establish an SSH connection.

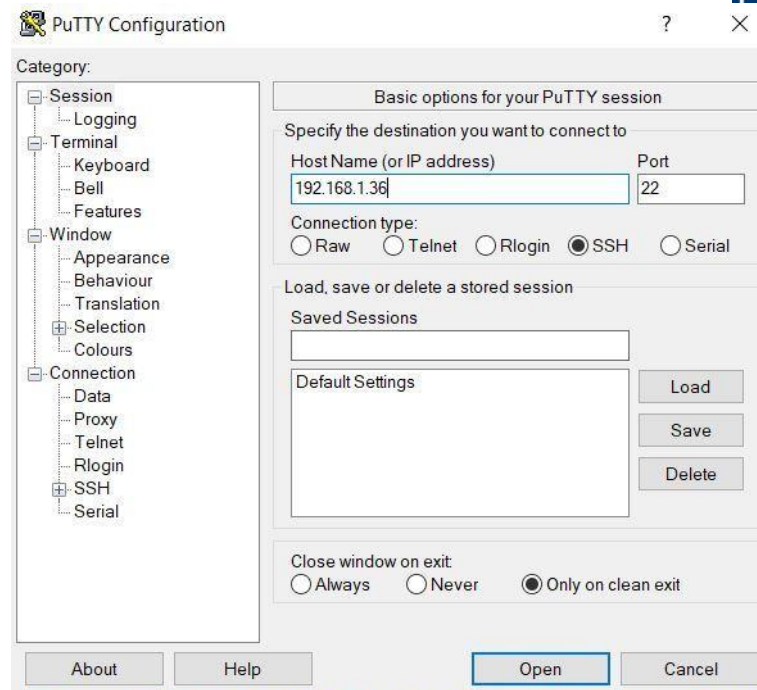
- [Download and install Putty](#) if you don't already have it. Putty is the leading SSH client for Windows.



- Enter the IP address of Raspberry pi device you wish to connect to in



Putty,
and click
Open.



orrect IP address will ask you to add this as a secure connection, choose the appropriate option.

- This will open a terminal where you have to enter the username and password. By default it is `pi` and `raspberrypi`

```

pi@raspberrypi: ~
login as: pi
pi@192.168.1.36's password:
Linux raspberrypi 5.10.11-v7l+ #1399 SMP Thu Jan 28 12:09:48 GMT 2021 armv7l

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Feb 17 11:53:19 2021 from 192.168.1.14
pi@raspberrypi:~ $

```



- Now we should enable the vnc to use the raspberry pi through the laptop.
- Enter `sudo raspi-config` after logging in into raspberry pi. It will open a configuration application in the device.

```
pi@raspberrypi: ~  
Raspberry Pi 4 Model B Rev 1.4  
Raspberry Pi Software Configuration Tool (raspi-config)  
  
1 System Options          Configure system settings  
2 Display Options        Configure display settings  
3 Interface Options       Configure connections to peripherals  
4 Performance Options     Configure performance settings  
5 Localisation Options   Configure language and regional settings  
6 Advanced Options       Configure advanced settings  
8 Update                 Update this tool to the latest version  
9 About raspi-config     Information about this configuration tool  
  
<Select>                <Finish>
```

- Go to the Interface option with the help of arrow keys. Click Enter.

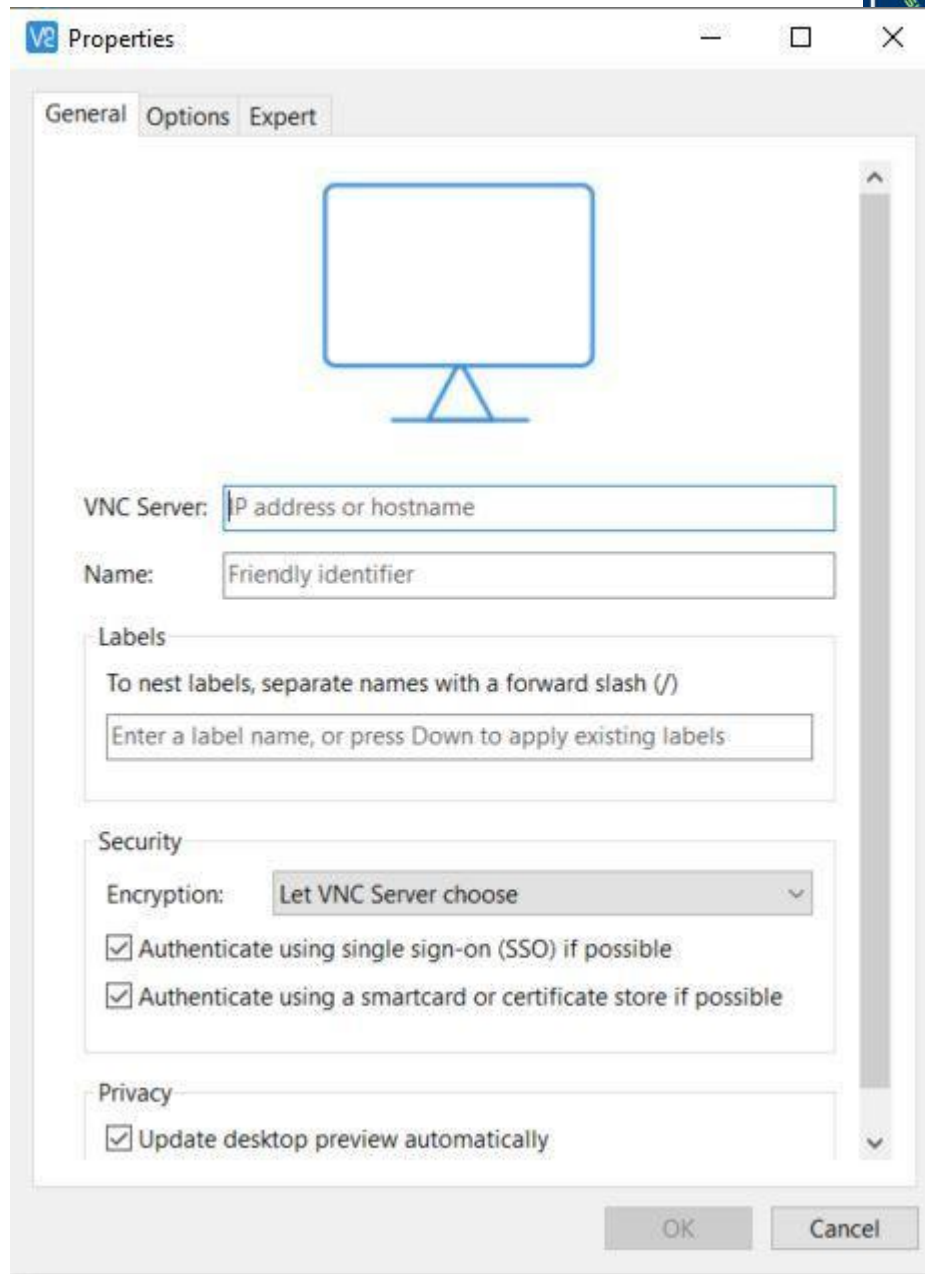
```
pi@raspberrypi: ~  
Raspberry Pi Software Configuration Tool (raspi-config)  
  
P1 Camera      Enable/disable connection to the Raspberry Pi Camera  
P2 SSH         Enable/disable remote command line access using SSH  
P3 VNC         Enable/disable graphical remote access using RealVNC  
P4 SPI         Enable/disable automatic loading of SPI kernel module  
P5 I2C         Enable/disable automatic loading of I2C kernel module  
P6 Serial Port Enable/disable shell messages on the serial connection  
P7 1-Wire      Enable/disable one-wire interface  
P8 Remote GPIO Enable/disable remote access to GPIO pins  
  
<Select>                <Back>
```

- On this page choose VNC and make sure that this service is enabled.
- Now install the [vnc viewer](#) on your computer. Choose the correct OS windows/mac/ubuntu.

- Once installation is done. Open the vnc viewer and click on file and choose

New
Connectio
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Provide the IP address of the Raspberry Pi IP address and enter the name as an identifier to recognize the device.

- Click on OK. Correct IP address will ask you to add this as a secure connection. Choose the valid option here.
- Now the next page will look as mentioned below:

V2 Authentication

 **Authenticate to VNC Server**
192.168.1.36::5900 (TCP)

Enter VNC Server credentials
(Hint: NOT your RealVNC account details)

Username:

Password:

☐ Remember password [Forgot password](#)

Catchphrase: Monaco Andy Norway. Corner pretty ma

Signature: ee-6c-4d-96-4e-b7-1e-43

- Enter the default username and password as pi and raspberry respectively.
- Now once you should be able to access the raspberry pi GUI. First time login will ask you to complete the trivial setup such as language, timezone, keyboard setup.

Temperature Sensor LM75A

The LM75A is an industry-standard digital temperature sensor with an integrated sigma-delta analog-to-digital converter (ADC) and I2C interface. The LM75A provides 9-bit digital temperature readings with an accuracy of $\pm 2^{\circ}\text{C}$ from -25°C to 100°C and $\pm 3^{\circ}\text{C}$ over -55°C to 125°C .





The LM75A operates with a single supply from 2.7 V to 5.5 V. Communication is accomplished over a 2-wire interface which operates up to 400 kHz. The LM75A has three address pins, allowing up to eight LM75A devices to operate on the same 2-wire bus. The LM75A has a dedicated over temperature output (O.S.) with programmable limit and hysteresis. This output has programmable fault tolerance, which lets the user to define the number of consecutive error conditions that must occur before O.S. is activated. The wide temperature and supply range and I2C interface make the LM75A ideal for a number of applications including base stations, electronic test equipment, office electronics, personal computers, and any other system in which thermal management is critical to performance.

Features

- No External Components Required
- Shutdown Mode to Minimize Power Consumption
- Up to Eight LM75As can be Connected to a Single Bus
- Power up Defaults Permit Stand-Alone Operation as Thermostat
- Key Specifications:
 - Supply Voltage
 - LM75A: 2.7 V to 5.5 V
 - Supply Current
 - Operating: 280 μ A (Typical)
 - Shutdown: 4 μ A (Typical)
 - Temperature Accuracy
 - 25°C to 100°C: $\pm 2^\circ$ C (Max)
 - 55°C to 125°C: $\pm 3^\circ$ C (Max)

Applications

- General System Thermal Management
- Communications Infrastructure
- Electronic Test Equipment
- Environmental Monitoring

Documentation

- [LM75A datasheet](#)

Temperature Logger Experiment

In this experiment we are going to create a setup and circuit connection to sense the environment temperature. We will use raspberry pi and temperature sensors as the main hardware. Other essential hardware information is mentioned below.

The devices we are using in this experiment are easily available in the market.

Additional hardware requirement

- Jumper wire (Female to Female - 6)
- Breadboard (Optional)
- Raspberry Pi - 4
- LM75A temperature sensor

Circuit Connection

Please find the block diagram below to create the setup. We are directly connecting the device and sensor without using the breadboard. Please find the pin information of Raspberry Pi and circuit connection table below:



Circuit connection table

Raspberry Pi Pin	LM75A sensor pin
1	VCC
3	SDA
5	SLC
6	GND

There is another pin on the sensor named OS, it is to detect overheating of the sensor. Currently we will not leverage this feature.

Once the circuit is connected and jumper cables are connected we should write the script to sense the data.

Python Code

In order to use the LM75A sensor we need to install a few modules. Please follow the following steps.

- Open the terminal on your raspberry device.
- Enter the ``sudo apt-get update``
- Enter ``sudo apt-get install git build-essential python-dev python-smbus i2c-tools``
- After that type ``raspi-config`` and goto the Interface option and enable I2C
- Reboot the device to complete this step.

Now create a python file with the name of your choice and enter the following code. Take care of indentation while copying the code.

```
import sys
import smbus
import time

# By default the address of LM75A is set to 0x48
# aka A0, A1, and A2 are set to GND (0v).
address = 0x48

# Read I2C data and calculate temperature
count = 0
while True:
    bus = smbus.SMBus(1)
    raw = bus.read_word_data(address, 0) & 0xFFFF
    raw = ((raw << 8) & 0xFF00) + (raw >> 8)
    temperature = (raw / 32.0) / 8.0
```

```
# Print temperature
print('Temperature: {0:0.2f} *C'.format(temperature))
count += 1
time.sleep(1)
```



Output

Once the code is edited and circuit connections are correct, by running the following command you will be able to view the logged data on the terminal.

A screenshot of a terminal window on a Raspberry Pi. The window title is 'pi@raspberrypi ~'. The terminal shows the command 'python3 LM75Atemp.py' being executed. The output consists of ten lines of temperature readings in Celsius, formatted to two decimal places. The readings are: 34.88, 34.88, 34.88, 34.88, 34.75, 34.75, 34.88, 34.88, 34.88, and 34.75. The terminal has a menu bar with 'File', 'Edit', 'Tabs', and 'Help'.

```
pi@raspberrypi ~
File Edit Tabs Help
pi@raspberrypi:~ $ python3 LM75Atemp.py
Temperature: 34.88 *C
Temperature: 34.88 *C
Temperature: 34.88 *C
Temperature: 34.88 *C
Temperature: 34.75 *C
Temperature: 34.75 *C
Temperature: 34.88 *C
Temperature: 34.88 *C
Temperature: 34.88 *C
Temperature: 34.75 *C
```